

Recomputing Godfrey–Isgur: An Excursion into a 1985 Quark Model

GIModel reproduction project

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Abstract

Forty years after Godfrey and Isgur published their relativized quark model [Phys. Rev. D **32**, 189 (1985)], it is still the default $q\bar{q}$ baseline — and its numbers still cannot be regenerated from the paper alone. I set out to recompute them, and this report is an account of that attempt rather than a description of a finished tool. It follows the route I actually took: a finite-difference solver as a first foothold, then the long way back to the paper’s own harmonic-oscillator method and its three-stage order of diagonalization and mixing. Four corrections along the way — about normalization and phase, about a converged approximation that was nevertheless the wrong calculation, about what convergence does and does not certify, and about verification infrastructure that reported the wrong run — taught me more physics than the successful steps did. Much of the work was carried out with coding agents, and the false starts are part of the story. What the excursion left behind is a Julia package whose intermediate quantities can be inspected, a comparison package that keeps the provenance and the still-open residuals, and a course and set of notebooks for anyone who wants to walk the same road.

1 Why I went back to 1985

The relativized quark model of Godfrey and Isgur (Godfrey and Isgur 1985) — hereafter GI — describes light and heavy mesons, “from the pion to the upsilon,” within a single soft-QCD Hamiltonian. One fixed set of parameters produces the masses, and afterwards the couplings, of essentially the whole meson landscape. Four decades later it is still the baseline against which new states are judged.

What drew me in was the gap between that status and what one can actually do with the paper. The relativistic smearing, the momentum-dependent operator factors, and the isoscalar annihilation machinery are specified semiquantitatively, spread across the main text and Appendix A, and the spectra themselves appear as figures rather than tables. To use the model you must first rebuild it, and everyone who has rebuilt it has done so privately. I wanted to know whether the published numbers really do follow from the published ingredients, with no fitting of my own — and I wanted the answer to be inspectable afterwards, by me and by anyone else.

So this is a trip report. It follows the order in which

things happened: what the paper asks you to compute (Section 2), the route I took through it and why it was the wrong way round (Section 3), the four mistakes that taught me the most (Section 4), what it was like to do this with coding agents (Section 5), what the reproduction currently rests on and where it is still incomplete (Section 6), and what the excursion left behind for other people (Section 7). Throughout, the 1985 paper is the authority; my code is only a claim about it.

I make no claim to be the first person to program this model. The point of doing it in the open is narrower and, I think, more useful: this particular calculation now exists with its assumptions, its conventions, and its evidence attached to it.

2 What the paper asks you to compute

2.1 The Hamiltonian

GI start from a rest-frame equation with a relativistic kinetic term,

$$H|\Psi\rangle = (H_0 + V)|\Psi\rangle = E|\Psi\rangle, \quad (1)$$

$$H_0 = \sqrt{p^2 + m_1^2} + \sqrt{p^2 + m_2^2}, \quad (2)$$

with $V = V(\mathbf{p}, \mathbf{r})$ momentum dependent and $\mathbf{p} = \mathbf{p}_1 = -\mathbf{p}_2$. In the nonrelativistic limit V reduces to the familiar sum of confinement, hyperfine, and spin–orbit terms. The spin-independent piece combines a linear Lorentz-scalar term with a Coulomb-type one-gluon exchange,

$$H_{ij}^{\text{conf}} = -\left[\frac{3}{4}c + \frac{3}{4}br - \frac{\alpha_s(r)}{r}\right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (3)$$

with $\langle \mathbf{F}_i \cdot \mathbf{F}_j \rangle = -4/3$ in a meson; the color hyperfine interaction supplies the contact and tensor pieces,

$$H_{ij}^{\text{hyp}} = -\frac{\alpha_s(r)}{m_i m_j} \left[\frac{8\pi}{3} \mathbf{S}_i \cdot \mathbf{S}_j \delta^3(\mathbf{r}) + \frac{1}{r^3} S_{ij} \right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (4)$$

and the spin–orbit interaction splits into a color-magnetic and a Thomas-precession part. A separate gluon-annihilation term H_A contributes only in self-conjugate isoscalar channels.

As GI themselves stress, Equation 4 and its spin–orbit companions are more singular than r^{-2} and are illegal operators in Equation 1 as written. The resolution is Appendix A, and Appendix A is where the reproduction work actually lives.

2.2 Relativization: smearing and momentum factors

Two things happen there. First, the relative coordinate is *smeared* over distances of order the inverse quark masses,

$$\rho_{ij}(\mathbf{r}' - \mathbf{r}) = \frac{\sigma_{ij}^3}{\pi^{3/2}} e^{-\sigma_{ij}^2(\mathbf{r}' - \mathbf{r})^2}, \quad (5)$$

with a mass-dependent width $\sigma_{ij}(\sigma_0, s)$, which tames the short-distance singularities and makes the operators legal. Second, the coefficients acquire a dependence on the quark momenta: mass factors m^{-1} are promoted to operator factors built from $E_i = (p^2 + m_i^2)^{1/2}$, so that the light-quark hyperfine term saturates to a finite $m \rightarrow 0$ limit controlled by $\langle p^{-1} \rangle$ instead of diverging.

Concretely, the central Coulomb and confinement terms become their smeared forms $\tilde{G}(r)$ and $\tilde{S}(r)$, and the Coulomb operator is dressed with a Hermitian momentum sandwich,

$$G' = A(p) \tilde{G} A(p), \quad A(p) = \left(1 + \frac{p^2}{E_1 E_2}\right)^{1/2}, \quad (6)$$

with the same construction, at the side exponents $\frac{1}{2} + \epsilon_i$ of Table II, applied to the contact, tensor, and spin-orbit kernels. Those kernels are themselves built from derivatives of the *smeared* central functions. Getting the ordering of these operations right, and keeping it right in two different numerical representations, turned out to be most of the project.

The strong coupling is a frozen sum of Gaussians,

$$\alpha_s(Q^2) = 0.25 e^{-Q^2} + 0.15 e^{-Q^2/10} + 0.20 e^{-Q^2/1000}, \quad (7)$$

with Q in GeV and $\alpha_s^{\text{crit}} = 0.60$, which becomes a coordinate-space error-function profile convenient for convolving with Equation 5. In self-conjugate isoscalars, $q\bar{q}$ annihilates through gluons ($n = 2$ for $C = +$, $n = 3$ for $C = -$), and the channel-coupling element

$$A^{(2S+1)L_J}_{ji} = 4\pi(2L+1) A^{(2S+1)L_J} \times \left[\frac{\alpha_s(M_j^2) \alpha_s(M_i^2)}{\pi^2} \right]^{n/2} \frac{S_L(\Psi_j) S_L(\Psi_i)}{m_i m_j} \quad (8)$$

depends on the wavefunction near the origin through $S_L(\Psi_i)$. That dependence is what later made the isoscalars the sharpest test of whether I had the wavefunctions right, not merely the energies.

2.3 The inputs I allowed myself

The parameters are GI Table II, taken verbatim (Table~1). These constants, the α_s form above, and the paper's reference masses are the only inputs; nothing in the model is fitted here.

GI quote an expected accuracy of 25 MeV for light-quark mesons and 10 MeV for heavy-quark systems, dominated by neglected coupled-channel shifts and by the schematic

Table 1: GI Table II parameters, used verbatim. The tensor exponent is the paper's ϵ_f (code `epsilon_t`).

Quantity	Symbol	Value
Light quark mass	$\frac{1}{2}(m_u + m_d)$	220 MeV
Strange mass	m_s	419 MeV
Charm mass	m_c	1628 MeV
Bottom mass	m_b	4977 MeV
String tension	b	0.18 GeV ²
Constant	c	−253 MeV
Frozen coupling	α_s^{crit}	0.60
QCD scale	Λ	200 MeV
Smearing width	σ_0	1.80 GeV
Smearing exponent	s	1.55
Contact factor	ϵ_c	−0.168
Tensor factor	ϵ_f	+0.025
Vector spin-orbit	$\epsilon_{\text{so}(V)}$	−0.035
Scalar spin-orbit	$\epsilon_{\text{so}(S)}$	+0.055

relativization. That is the standard the reproduction has to be measured against — and it is worth saying early that a solver tolerance and a physical accuracy are different things, a distinction I had to keep re-learning.

3 The route I took, and why it was backwards

In 1985 the harmonic-oscillator (HO) basis was not a stylistic choice. It was the practical route through an operator that depends on both position and momentum: oscillator functions have tractable representations in both spaces, and Appendix A inserts intermediate states to assemble products of the corresponding matrix elements. The paper's procedure has three stages. The relativized Hamiltonian is diagonalized in a large HO basis within fixed $|jm; ls\rangle$ sectors, the basis being enlarged until convergence. The smaller mass matrices generated by the off-diagonal tensor (${}^3L_J \leftrightarrow {}^3L'_J$) and antisymmetric spin-orbit (${}^3L_J \leftrightarrow {}^1L_J$, unequal masses only) couplings are then diagonalized in the basis of those eigenvectors. Finally, self-conjugate isoscalars acquire the annihilation matrix.

I did not start there. A finite-difference (FD) radial calculation was the easier foothold: grid values, explicit operator matrices, and contribution plots gave me something I could look at directly, term by term, when nothing yet worked. That was the right first move for a first implementation, and the wrong final answer. Once the FD spectrum stood up, its independence became the lever for going back to native HO matrix elements, and then to the paper's own order of diagonalization and mixing. The sequence was FD to HO, with repeated comparison at every step — a lesson about how *this* project went, not a general ranking of the two methods.

The intermediate stage was the most instructive, because

it looked finished when it was not. Representing a wave in an HO expansion does not by itself reproduce the paper’s algorithm. Native operator assembly, full fixed-sector diagonalization, and the later mixing stages all still had to be recovered individually. The native HO path now evaluates position- and momentum-dependent factors in their own representations, following Appendix A, optimizes the oscillator scale, and enlarges the basis until every requested energy passes a stated convergence test — currently 0.1 MeV over two consecutive refinements, supplemented by wave-overlap diagnostics and wave-sensitive audits. FD survives as an independent comparator, which is a better fate than being the headline.

4 Four corrections that taught me physics

The useful part of the trip was not the code that worked. It was the code that worked *and was wrong*, and what it took to notice.

Normalization and relative phase. A diagonal Rayleigh quotient is unchanged when its input vector is rescaled; a bare off-diagonal overlap is not. So comparing sampled FD waves with reconstructed HO waves demanded a common physical normalization, and without it perfectly matching diagonal results happily coexisted with badly wrong mixing matrix elements. The fix is recorded in commit [a987862](#). The HO phase-convention correction in [9ff437b](#) is the same lesson from the other side: an individual eigenvector has an arbitrary phase, but every operator and every mixed-state composition must transform consistently with whatever phase you chose. Both are statements about physics that I met as implementation failures.

A converged approximation can still be the wrong calculation. A nonperturbative solve that was unavailable in one path had been quietly falling back to a first-order shift. Commit [eef789b](#) made that case fail loudly instead. The later fixed-sector work ([b9d3474](#)) put the full diagonal spin interactions into the Hamiltonian *before* solving it, rather than adding them afterwards. Recovering the masses was not enough: the spin-distorted wavefunctions have to reach the decay and annihilation observables too, which is exactly where a first-order shortcut stops being invisible. The completion in [78a3d76](#) also removed extra spin-strength bridge factors once the literal Appendix-A coefficients were correct.

Convergence certifies less than it appears to. Commit [8655d68](#) removed the mesh from the HO Appendix-A assembly, and [39ee9c8](#) made adaptive basis refinement a production requirement rather than a diagnostic. But energy convergence had to be complemented by tests of origin-sensitive functionals and of the final mixed-state observables — an energy can be stable to 0.1 MeV while $S_L(\Psi)$ in Equation 8 is still moving. The independent FD certification in [bdd65d8](#) gave a second numerical route

with its own spacing and domain checks, but agreement between two routes only became evidence about the physics once I could say precisely which representations and which approximations each of them was making.

Verification infrastructure can hide the mistake. Commit [959fdb0](#) corrected a scorecard that was faithfully reporting the *previous* verification run, and [2fd3fd7](#) restored FD/HO audits that had drifted outside the gate. This is not a clerical detail. A correct-looking report that does not describe the code you are inspecting is worse than no report, and keeping the reference comparison and its generation path explicit is part of the physics, not housekeeping around it.

5 Doing this with agents

Much of the work was carried out with coding agents, and the git history records the whole shape of it: extraction from the paper, implementation, autonomous iterations, physics audits, repairs, and eventually teaching material. The paper, the numerical checks, and my own reading supplied the criteria for accepting any of it.

The honest summary is that the failure modes above are the story. Plausible code, passing local tests, and a reassuring residual could all coexist with a missing physical operation. What made the difference was the corrective loop: a discrepancy raised a question about normalization, or operator staging, or provenance; the question sent me back to the paper or to an independent calculation; the answer changed both the code and my explanation of it. That loop is also why the mistakes are worth writing down — each one is a small lesson in quantum mechanics that happens to be reachable through a diff.

These are episodes from one repository, not a controlled measurement of agent productivity or reliability. The commit identifiers here are navigation anchors: the changes and their audits are the evidence, not the commit titles.

6 What the reproduction rests on, and where it stops

GIPaper keeps a manifest of paper units with links to code, tests, reference sources, and residual reports — the detailed record of what has actually been computed and compared. Reference data keep their origin, whether a printed table or a digitized figure. Parameter provenance is tracked with the same care, because taking the paper’s inputs, repeating a fit the paper performed, and calibrating a diagnostic control against reference targets are three different operations, and only the first two are predictions. The reports label which is which so that a control cannot quietly become evidence.

One compact example of the current state: the photon audit gives $\mu/\mu_N = +0.688$ for $\rho \rightarrow \pi\gamma$ against the printed $+0.69$, and -1.200 for $K^{*0} \rightarrow K^0\gamma$ against -1.20 . Those

are values from the checked-in audit of September 7, 2026, not a fresh calculation made for this manuscript. The ρ row is also the row GI used to fit a relativization exponent, so its agreement tests the reconstructed pipeline rather than supplying an independent prediction — exactly the kind of distinction the provenance labels exist to preserve. Other transitions probe different flavor weights, wavefunctions, and cancellations, and are the more interesting test.

The aim is full-depth reproduction, and the repository audit is candid that it is not there yet: complete accounting of the Table VI photon-decay rows and the characterization of their residuals are still open, and some mixing- and cancellation-sensitive quantities retain visible discrepancies. So this report does not claim exact agreement for every printed number, nor whole-paper coverage. The native three-stage spectrum implementation and the independent FD comparator are done and documented separately from that remaining work; the boundary should move as the audits close, and the audit files, not this report, are where it is authoritative.

7 What the excursion left behind

7.1 A calculation you can take apart

The thing I most wanted at the start was the thing that did not exist: a version of this calculation whose intermediate quantities you can look at. In the model a rest-frame mass is an eigenvalue of the Hamiltonian, whose kinetic part is Equation 2 and whose remainder is the confinement and spin-dependent structure of Section 2. When configurations mix, a physical state is

$$|\Psi\rangle = \sum_a c_a |\psi_a\rangle,$$

with signed coefficients and explicit flavor content, and a transition is a coherent sum,

$$\langle\Psi_f|O|\Psi_i\rangle = \sum_{a,b} c_{f,a}^* c_{i,b} \langle\psi_{f,a}|O|\psi_{i,b}\rangle.$$

Writing it out is the quickest way to see why a mixing *probability* cannot predict a decay: the relative phases can enhance or cancel. A width needs the decay operator, the normalization, and the kinematics as well; the mass spectrum alone does not determine it.

The package exposes those stages in that order. A spectrum contains identifiable states; a state resolves into physical components; components lead to solver-native waves; waves lead to radial matrix elements; the comparison layer relates the result to a printed number. What stays visible throughout is which quantity is an input, which is computed, and which is a reference value — the three things I most often confused with each other while getting here.

This also makes conversational use of the model reasonable: a language model can turn a physics question into a package calculation, explain the objects that come back, and point

at the methods involved, while the numbers come from executing the code and carry their parameters, labels, solver settings, and convergence information with them. That is an intended use of the existing interface, not a claim that a validated conversational product exists.

7.2 A course, and notebooks

Somewhere along the way the reproduction turned into teaching material, because the questions I had to answer for the code were the questions a student has anyway. The theory course is nine sheets: radial quantum mechanics, angular momentum and meson quantum numbers, variational and semirelativistic methods, confinement and the running coupling, relativization and smearing, spin fine structure, mixing, decay observables, and the complete spectrum algorithm. Every problem has a worked solution and a short concept check. A computational-discovery sheet starts the bridge to numerical work; the full hands-on course is still ahead.

Each piece has its own job. The paper gives the original model and its reasoning. The course slows the derivations down and makes you do them. The notebooks make the assumptions adjustable and the consequences immediate. This report is the map, and deliberately does not repeat the course.

Pluto suits the last of those: executable cells, reactive dependencies, mathematical prose, controls, and results that redraw when you change a mass or a solver setting. Live documentation and linked docstrings make a notebook an entry point into the package rather than a demonstration beside it, and the documentation graph links the public operations explicitly, with a static link audit checking that the path from any explanation to the next numerical question actually exists. The adaptive HO notebook turns the convergence question of Section 3 into the lesson: it follows the oscillator scale and the basis refinements, shows the energies settling, and asks the reader why a curve that looks flat still needs a criterion — and why energy convergence does not certify every wave-sensitive observable.

7.3 Seeing the model move

One consequence of an executable model surprised me by how much it explains. The heavy-quark notebook varies a constituent heavy-quark mass continuously while the shared interaction parameters stay fixed, which connects the model's D and B sectors, or its D_s and B_s sectors, or — varying both masses — charmonium and bottomonium. Subtracting the constituent rest masses separates the change in binding and excitation structure from the plain rise in total mass, and hyperfine splittings, level ordering, and the response of the light degrees of freedom become things you watch rather than tabulate.

Two honest caveats belong with that picture. The heavy masses are external parameters of a constituent model with a shared phenomenological interaction; the interpolation is

a family of model calculations, not a physical decay of a bottom meson into a charmed one. And the same access to wavefunctions invites structure pictures — radial densities, nodes, characteristic radii, angular distributions — which are only interpretable if they say what they display: for a normalized reduced radial wave $|u(r)|^2 dr$ is a radial probability, while the three-dimensional density carries the angular factor too, and mixed states need their coherent composition and the right spin or flavor sums. The ingredients for a curated meson-structure animation are all there; the animation itself is not yet.

8 Where I would go next

The obvious continuations are physics rather than software. Screened interactions (Li and Chao 2009) and explicit coupled-channel treatments (Barnes and Swanson 2008; Ortega et al. 2010) are the two directions in which the GI baseline has mostly been extended, and each changes the model and needs its own validation. Closer to hand, the paper’s quark-emission operator could be applied directly to the calculated physical wavefunctions instead of the single-oscillator-scale approximation used for its numerical strong-decay tables — which would expose the role of radial nodes, state-dependent scales, and mixing in those amplitudes. That is the test I most want to run, because it uses the wavefunctions rather than only the eigenvalues, and the wavefunctions are what this whole excursion was really about.

Julia makes one more direction tempting. For an isolated normalized eigenstate,

$$\frac{\partial E_n}{\partial \lambda} = \left\langle \psi_n \left| \frac{\partial H}{\partial \lambda} \right| \psi_n \right\rangle$$

is an explicit starting point for sensitivity studies and fitting. The repository already contains the numerical-type preparation and FD and fixed-scale HO derivative probes. End-to-end automatic differentiation is future work, and not trivially so: adaptive basis selection, level crossings, and derivatives of mixed eigenvectors all need care. Whatever comes out of it will need the same checks the original computation did.

If there is a moral to the trip, it is the one from Section 4: the model became clear to me at exactly the points where my code was confidently wrong. That is the experience I would like the course and the notebooks to make available to someone else — ask a question, get a number, take the number apart, and find out what you had assumed.

A Repository reading guide

Where the supporting material lives.

- **Learning:** `LearningTrack/README.md` and `examples/README.md` introduce the theory sheets and computational notebooks.

- **Discoverability:** `docs/discoverability_graph.md` records the linked public API; `docs/formula_map.md` connects implementation to paper equations.
- **Reproduction:** `docs/paper_manifest/` and `docs/paper_dashboard.qmd` provide the per-unit inventory. `docs/reproduction_audit.md` records the cross-cutting interpretation and remaining scope.
- **Numerical evidence:** `GIPaper/docs/residual_reports/` contains the HO convergence, FD comparator, spectrum, mixing, and decay audits. The photon numbers quoted in Section 6 come from `table_vi_photon_decays.md` there.
- **Development:** `docs/original_1985_algorithm_audit.md`, `docs/incident_followup_audit.md`, and the git history document the recovery of the paper’s algorithm and the corrections of Section 4.
- **Differentiation:** `diff_support/implementation_status.md` distinguishes completed preparation and probes from proposed derivative capabilities.

References

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